Priyanshu Ranjan Gupta

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EDUCATION

Doctor of Philosophy[1] in Civil Engineering (Environmental and Water Resources Engineering) *2022 - Present*
University of Texas at Austin, USA *CGPA: Ongoing*

Master of Technology in Chemical Engineering [Rank 2 in class of 89] *2020-2022*
Indian Institute of Technology, Bombay *CGPA: 9.56/10*

Bachelor of Technology (with Honours) in Chemical Engineering *2015-2019*
Indian Institute of Technology, Gandhinagar *CGPA: 8.13/10*

PUBLICATIONS

- Yao, C., Ben-Zvi, A. M., Xu, R., Ram, N., Stolov, M., **Gupta, P.R.**, ... & Segal-Peretz, T. (2025). **3D Nanoscale Structures of Hydrated Polyamide Desalination Membranes Revealed by Cryogenic Transmission Electron Microscopy Tomography.* ACS nano, 19(17), 16718-16731. https://doi.org/10.1021/acsnano.5c01190
  *This paper was selected for cover art
- Patel, C. G., **Gupta, P. R.**, & Swaminathan, J. (2024). *Saline Brine Separation: Thermodynamic and Techno-Economic Comparison of Counterflow Reverse Osmosis with Electrodialysis.* Industrial & Engineering Chemistry Research, 63(38), 16509-16520. https://doi.org/10.1021/acs.iecr.4c01644
- **Gupta, P.R.**, Shanmukham, S.P., Patel, C., Lienhard, J. H., Swaminathan, J. (2022). *Replacing chloride anions in dyeing enables cheaper effluent concentration and recycling* Desalination, Volume 533, 2022. https://doi.org/10.1016/j.desal.2022.115761.
  Pre-print Version- Link

CONFERENCE POSTER PRESENTATIONS

- ## **Gupta, P. R.***, Nance, H. T., Nguyen, K., Srivathankul, M., Tang, E., Deegan, J. C., Dhiman, R., Carr, J., & Kumar, M. (2025, May). *Identifying industrial predictors of PFAS violations in U.S. public water systems .* **Association of Environmental Engineering and Science Professors (AEESP) Conference 2025**, Duke University, Durham, NC, United States.
  ##: Awarded Best Poster Award (32 out of 450)
- Doltkhah A*, **Gupta PR**, Wilson LD, Adjustable catalytic activity in redox-active polymer for selective organic dyes reduction. Poster Presented at: 103^{rd} **Canadian Chemistry Conference and Exhibition**; 2020 May 28; Winnipeg, MB, Canada
- **Gupta P.R.***, Doltkhah A, Wilson LD, Supported Silver Nanoparticles (AgNP) for Catalytic Reduction and Adsorption Processes. Poster Presented at: **AIChE Student Regional Conference**; 2018 August 11; VIT Vellore, India
- **Gupta P.R.***, Doltkhah A, Wilson LD, Supported Silver Nanoparticles (AgNP) for Catalytic Reduction and Adsorption Processes. Poster Presented at: **Undergraduate Research Conclave (UGRC)**; 2018 August 25; IIT Gandhinagar

AWARDS AND SCHOLARSHIPS

- **Best Poster Award**, *Association of Environmental Engineering & Science Professors* *2025*
- **Mentor for the First Placed Team**, *2024 Green Hackathon @ Maseeh Department of CAEE, UT Austin (Link)* *2024*
- **F.L.Y. (Finding the Leader in You) Scholar**, *Competitiveness Mindset Institute, USA* *2021*
- **TATA Fellowship (2020-22)**, *TATA Center for Technology and Design (TCTD), IIT Bombay (Link)* *2020*
- **Dean's List Award for Academic Excellence**, *IIT Gandhinagar* *2019*
- **Special Mention for Poster Presentation**, *Undergraduate Research Conclave, IIT Gandhinagar* *2018*
- **Dean's List Award for Academic Excellence**, *IIT Gandhinagar* *2017*

[1] Ongoing. Expected Graduation: July 2027

RESEARCH EXPERIENCE

Machine learning driven identification statistical association between industrial activity type and PFAS violation in U.S. Public Water Systems (PWS) *Oct 2024-Ongoing*
Dr. Manish Kumar, UT Austin

- Engineered a feature pipeline to generate 35-dimensional descriptors for each U.S. HUC8 region with a PWS.
- Trained a 10-fold balanced random forest model predicting PFAS detection at the HUC8 level | Achieved an ROC-AUC of 0.8, and f1-score of 0.74.
- Identified plastics manufacturing and metal & electroplating industries as key contributors to PFAS violations, guiding future research.

Machine learning driven classification of ion binding pockets in proteins *Sep 2022-Ongoing*
Dr. Manish Kumar, UT Austin

- Developed a computational workflow for extracting and embedding protein-ion interaction sites using ESM-2-based vector representations for all available protein crystal structures with bound ions.
- Trained a supervised neural network classifier to predict ion selectivity from these embeddings, achieving an accuracy of 96% and an ROC-AUC score of 0.91.
- Currently designing an end-to-end pipeline to fine-tune the classifier for integration with a generative model aimed at engineering de novo ion-selective protein channels, with potential applications in selective ion-ion separation.

Development of a Rotary Dryer for the Jaggery Industry *Aug 2021-Jun 2022*
M. Tech. Thesis Project | Dr. Sanjay Mahajani, IIT Bombay
(Sponsored by Tata Centre for Technology and Design)

- Conducted **extensive literature survey** on techniques used to study particle motion in rotating drum
- Developed computational model in **ANSYS Fluent** using Eulerian-Eulerian approach and in LIGGGHTS using **Discrete Element Method** to study distribution of powder particles between active passive phases in a flighted drum
- Mentored 4 undergraduate students to perform field experiments at the pilot plant in Kolhapur, MH, India

Gas exchange in lungs of COVID-19 infected patients *Apr-Sep 2021*
Independent Project | Dr. Sanjay Mahajani and Dr. Santosh Noronha, IIT Bombay

- Utilized the **Finite Difference Approach** to simulate the gas transfer inside a human lung using **Weibel geometry**
- Coded different **numerical solvers Euler, RK-4, Adams-Bashforth**) to solve the convection diffusion equation
- **Confirmed** the presence of diffusion dominated regime by showing $\mathbf{Pe} < \mathbf{1}$ in generations 17-23 of the bronchi

Zero-Liqud Discharge in Textile Dyeing Units *Feb-Aug 2020*
Junior Research Fellow | Dr. Jaichander Swaminathan, IIT Gandinagar

- **Analyzed** past experiment data to quantify the impact of process chemistry on dyeing quality using $\Delta E_{cmc,2:1}$
- **Designed** additional experiments for industrial partner to study effect of dyebath salinity on shade of dyed fabric
- **First author** in the resulting publication in the **Desalination** journal

Entropy Generation Minimization (EGM) in Brine Splitting technologies *Feb-Aug 2020*
Junior Research Fellow | Dr. Jaichander Swaminathan, IIT Gandinagar

- Implemented the EGM approach for osmotically assisted reverse osmosis **(OARO)** and Electrodialysis **(ED)** using *fsolve* and *ode45* functions in **MATLAB**
- Studied the impact of **concentration polarization and membrane properties** on specific energy consumption
- **Co-author** in a soon to be published journal article documenting the derivations, methods and results

Supported Silver Nanoparticles (AgNP) for Catalytic Reduction and Adsorption Processes *May-Jul 2018*
Visiting Research Scholar | Dr. Lee Wilson, University of Saskatchewan

- Used **in-situ UV-Vis studies** to investigate the performance of the AgNP catalysts supported by a customized polymer backbone
- Tuned the composition of polymer support to **reduce** the reaction time **by 50%** to <7minutes for reduction of methylene blue dye
- Collected compelling **evidence** to highlight role of hydrophobicity of support polymer in the performance of catalyst

TEACHING EXPERIENCE

Graduate Teaching Assistant *Aug 2024 - Dec 2024*
EVE 310 - Sustainable Systems Engineering, Fall 2024, UT Austin

- Designed and delivered hour long python tutorials to a class of 60 sophomore year Environmental Engineering undergraduates every week for the semester
- Covered topics like data manipulation, plotting, linear & logistic regression, and model evaluation using libraries like NumPy, Scikit-learn, and Pandas.
- Supervised, reviewed, graded, and provided feedback to the students on weekly assignments, mid-term exams, and final projects.
- Created two final project challenges for the class, one of which grew into a comprehensive research project undertaken by a team of six undergraduate students.

INDUSTRIAL EXPERIENCE

Management Trainee *Jun 2019 - Jan 2020*
Operations Excellence (OE) Team, Aarti Industries Limited

- **Inspected** and **identified gaps** in the existing **P&IDs** of Sulphuric Acid, CSA and DMS plants
- Examined performance metrics for **5 business functions** | Supported **leadership** in managing the division
- Monitored operational productivity for **3 manufacturing plants** | Ensured >**90%** overall equipment efficiency

SELECT TECHNICAL COURSE PROJECTS

Fenton Treatment – Batch to Continuous *Jan-Apr 2021*
Industry Defined Problem, SLP | Dr. Sanjay Mahajani, IIT Bombay
Business Model Canvas (BMC) for Automated Jaggery Manufacturing Units *Jan-Apr 2021*
Technology Design and Development Laboratory | Dr. Santosh Noronha, IIT Bombay
Design of a seawater desalination plant for producing potable water *Aug-Nov 2019*
Process Synthesis and Design | Dr. Babji Srinivasan, IIT Gandhinagar
Feasibility Study for setting up a P.E.T. bottle manufacturing plant *Mar-Apr 2018*
Process Plant Design – How to Set Up a Process Industry | Dr. Surya Pratap Mehrotra, IIT Gandhinagar
Synthesis of Curcumin nanoparticles *Jan-Apr 2018*
Project Course | Dr. Sameer V Dalvi, IIT Gandhinagar
Enzymatic production of Formaldehyde and Hydrogen Peroxide from methanol *Feb-Apr 2018*
Process Analysis and Simulation | Dr. Nitin Padhiyar, IIT Gandhinagar

GRADUATE LEVEL ELECTIVES COMPLETED

At IITB: Advanced Process synthesis, Product Research and Design, Granular Mechanics, Supervized Learning Project

At IITGn: Energy Efficient Design of Separation Processes, Energy Systems, Process Plant Design, Biological Physics

At UT Austin: Physical and Chemical Water Treatment, Water Pollution Chemistry, Mass Transfer in Polymers

VOLUNTARY SERVICE

President, *AIChE IIT Gandhinagar Student Chapter* *Aug 2018-Jul 2019*
Founder Organiser, *Drone Racing Amalthea, IIT Gandhinagar* *Aug-Oct 2017*
Instructor, *Multiple workshops on Arduino, MATLAB*
Volunteer, *Amalthea, IIT Gandhinagar* *2017*

References available on request